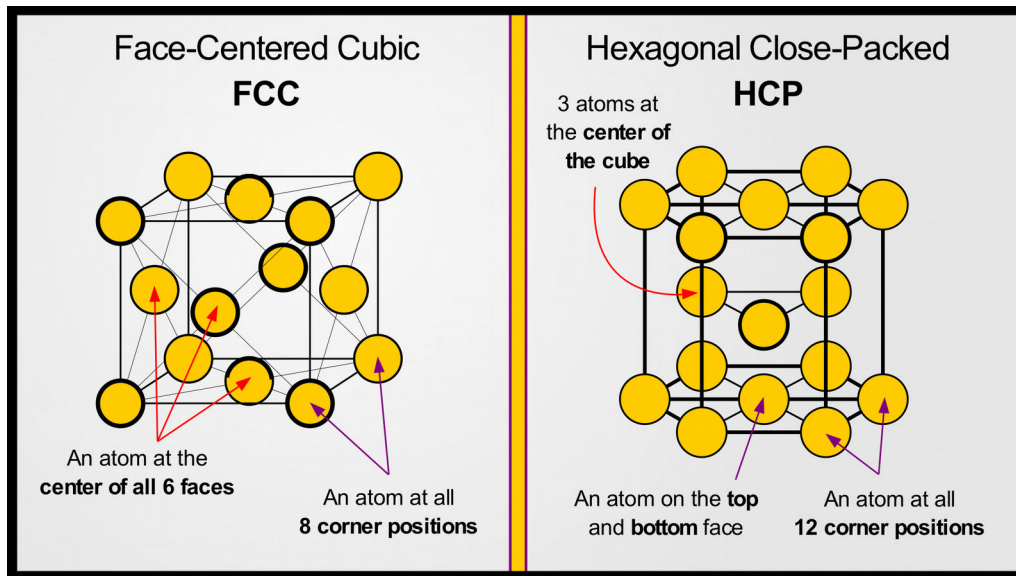


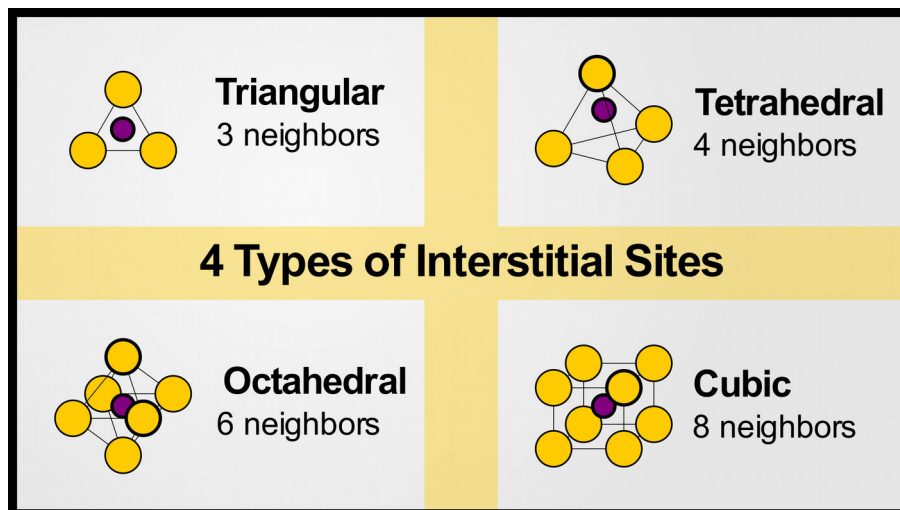
APS110 – Lecture 11

See hand written notes for drawings.



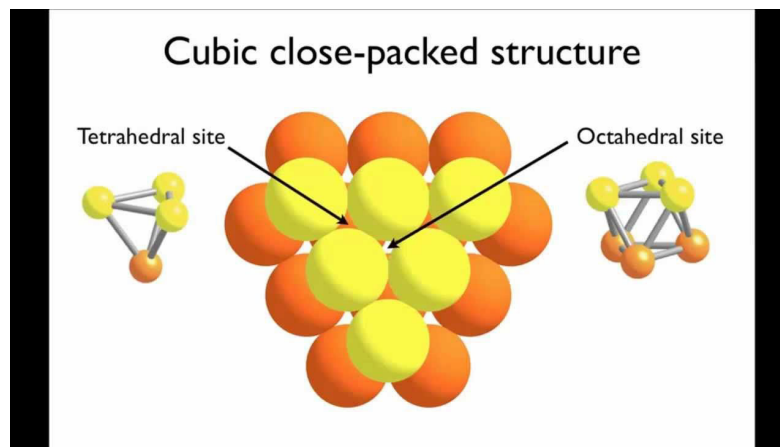
Octahedral Interstitial Sites

- We see this interstitial site in rock salt, with the position of the cation right in the center of the cube
 - $\text{CN}_{\text{cation}} = 6$
- We also have other interstitial sites $\rightarrow$ tetrahedral, simple cubic



- Note that the sizes of these interstitial sites differ – cubic has the largest interstitial site by far, then octahedral is smaller, then tetrahedral is even smaller
- In the case of rock salt (ceramic structures in general), the cations and anions are not necessarily going to be in contact – the atoms will be pushing each other apart

- We might thus want to calculate the “ideal size” of the interstitial site – what size of cation can fit perfectly in the interstitial site?
- You want to make an initial assessment of the sizes of the ions, so you can figure what the correct answer is likely going to be → in this case you can tell the cation is much smaller than the neighbor anions
 - It's likely that $R_{\text{cation}}/R_{\text{anion}}$ is around 0.5
 - Cation to anion atomic radius ratio
- “Geometrically ideal” → anions still touching, cation touching anions
- Depending on the R_c/R_A ratio, you'll understand what type of interstitial site will form with that crystal
 - If the size of the cation is really small in comparison to the anions, it will only really be able to fit in tetrahedral
 - If the cation is big enough to occupy octahedral, it will
 - And, if its big enough to occupy simple cubic, it will
- **Anytime you stack up close-packed planes, you'll create both octahedral and tetrahedral interstitial sites** → therefore, both FCC and HCP have empty octahedral and tetrahedral sites



Octahedral Site in FCC An interstitial site (purple dot) among neighboring 6 host atoms (yellow circles)	Octahedral Site in HCP An interstitial site (purple dot) among neighboring 6 host atoms (yellow circles)
Tetrahedral Site in FCC An interstitial site (purple dot) among neighboring 4 host atoms (yellow circles)	Tetrahedral Site in HCP An interstitial site (purple dot) among neighboring 4 host atoms (yellow circles)

Moving back to mechanical behaviour

- Which will have a higher Young's modulus?